

UMUT-II HYDROPOWER PLANT, TURKEY



Nisan Elektromekanik Enerji San. ve Tic. A.Ş.
www.nisanenerji.com

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1 PROJECT DETAILS

1.1 Summary Description of the Project

“Umut-II Hydropower Plant, Turkey” (hereafter referred to as the “Project”) owned by Nisan Elektromekanik Enerji San. ve Tic. A.Ş. (hereafter referred to as the “Project owner”) is a large scale, run-of-river hydropower project located on the Karakus River, Ordu, Turkey.

The Project owner is a private entity owned by Atac Insaat and Boydak Enerji. The main objective of the Project owner is to increase the power generating capacity in Turkey by developing, constructing and operating renewable energy power plants with the support of carbon credits.

The purpose of the Project is to generate electrical power in an efficient, clean, reliable and sustainable way with utmost respect on social and environmental aspects and to reduce emissions by partially substituting the electricity supply of fossil fuel fired power plants in Turkey.

The construction of the Project has started on 01.10.2010 and the generation is expected to start on 01.10.2012. The Project has estimated emission reductions of 37,883 t_{CO2e} annually.

1.2 Sectoral Scope and Project Type

Scope number 1: Energy industries (renewable - / non-renewable sources)

Project type: Grid-connected electricity generation from renewable sources

1.3 Project Proponent

Nisan Elektromekanik Enerji San. ve Tic. A.Ş. is the Project owner, as the Project proponent. The contact information is provided below;

<i>Name of Party involved (*) ((host) indicates a host Party)</i>	<i>Private and/or public entity(ies) project participants (*) (as applicable)</i>	<i>Kindly indicate if the Party involved wishes to be considered as project participant (Yes/No)</i>
Turkey (host)	Nisan Elektromekanik Enerji San. ve Tic. A.Ş.	No

Organization :	Nisan Elektromekanik Enerji San. ve Tic. A.Ş.
Street/P.O.Box:	Cinnah Caddesi. Ahenk Sokak. No:5 / 2 Çankaya
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City:	ANKARA
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1.4 Other Entities Involved in the Project

Mavi Consultants acts as carbon consultant for this Project. The contact information is provided below;

Organization:	Mavi Sürdürülebilir Kalkınma Proje ve Danışmanlık Hizmetleri Ltd. Sti.
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Last Name:	Capalov
Middle Name:	
First Name:	Lale
Department:	
Mobile:	
Direct FAX:	
Direct tel:	
Personal E-Mail:	

1.5 Project Start Date

According to VCS v.3, the project start date is defined as the date on which “the project began reducing or removing GHG emissions”¹. The Project is expected to start generating electricity as of 01.10.2012, which is the Project start date for this Project activity.

1.6 Project Crediting Period

For this type of project activities, VCS allows a maximum crediting period of 10 years, with the possibility to renew it twice². Therefore, the Project crediting period starts on 01.10.2012 and lasts

¹ Source: VCS Standard, Section 3.8 (<http://www.v-c-s.org/sites/v-c-s.org/files/VCS%20Standard%2C%20v3.0.pdf>)

² Source: VCS Web Page (<http://www.v-c-s.org/faqs/what-project-crediting-period-under-vcs-program>)

10 years.

1.7 Project Scale and Estimated GHG Emission Reductions or Removals

Projects that generate emission reductions more than 1,000,000 tCO₂/a are defined as mega projects in the VCS v.3³. This is not applicable for the Project activity.

Project	x
Mega-project	

Years	Estimated GHG emission reductions or removals (tCO ₂ e)
2012 (Partial)	9,860
2013	37,883
2014	37,883
2015	37,883
2016	37,883
2017	37,883
2018	37,883
2019	37,883
2020	37,883
2021	37,883
2022 (Partial)	28,023
Total estimated ERs	378,831
Total number of crediting years	10
Average annual ERs	37,883

Actual annual emission reductions can deviate from estimated figures and will be calculated based on actual net energy supplied to the grid according to the measurement device readings, as specified in the monitoring plan.

1.8 Description of the Project Activity

The Project involves the generation of electricity and sales of carbon credits. It will help the host country as a developing country to stimulate and commercialise the use of grid connected renewable energy technologies and markets.

It will make use of water from the Umut-II river tributaries. The Project employs 3 Francis-type turbines with 8.15 MW_e capacity each, totalling to 24.45 MW⁴. The estimated net power supply to the grid is 68,740 MWh per annum⁵. The Project will be operated all year round, at reduced capacity in the summer as a result of diminishing water flow.

³ Source: <http://www.v-c-s.org/docs/VCS%20Standard%20-%20v3.0.pdf>, Section 3.10

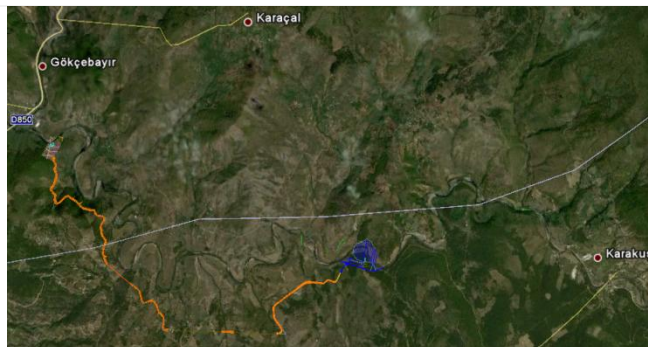
⁴ Source: Turbine order contract.

⁵ Source: Feasibility Study, Section 1.3.

The pressure of the water flow will drive the turbine blades, which will drive the coupled generators to produce electricity. The generated power will be used for internal consumption and the rest will be supplied to the national power grid. The generation license has been obtained on 01.05.2008 for 49 years, and the Project owner also foresees a lifetime of roughly 49 years, 0 months.

The Project will partially divert the water flow of Karakus River and its tributaries to multiple diversion weirs to regulate the water flow and convey it to the power house.

The Project has the sole purpose of energy generation and is a run-of-river type hydropower plant that will utilize a gross head of 152 m with a project flow of 18 m³/sec⁶.



All publicly and privately owned power plants are connected to the same national grid in the host country. Transmission line operator TEIAS manages the stability and quality of energy supply in the grid through its settlement body, the Electricity Market Financial Settlement Center⁷ (PMUM). PMUM regulates and manages electricity supply of the grid per power plant and also performs financial settlement activities. When the Project activity generates electricity, PMUM will ensure that the pool of other power plants generate less energy in order to regulate the overall grid supply. This means that as long as the Project feeds the grid with its clean energy, thermal power plants connected to the grid will operate less, consuming less fossil fuel and generating less CO₂ emissions at the extent that the Project supplies electricity to the grid that would be generated by the pool of other existing power plants in the absence of the Project.

1.9 Project Location

The Project is located in the Black Sea region of Turkey, near Akkus town, Ordu Province. The Power House is located close to the Gokcebayir village and the diversion weirs are located close to the Karakus village. The locations of the diversion weir and the power house are provided below:

Table 1. GPS Coordinates

Coordinate System	Diversion Weirs	Power House
Degrees, Minutes, Seconds (Latitude)	40° 42' 05"	40° 42' 46"
Degrees, Minutes, Seconds (Longitude)	37° 4' 12"	37° 1' 44"
Decimal (Latitude)	40.701389	40.712778
Decimal(Longitude)	37.07	37.028889

⁶ Source: Feasibility Study, Section 1.3.

⁷ Internet link: <http://dqpys.teias.gov.tr/dqpys>

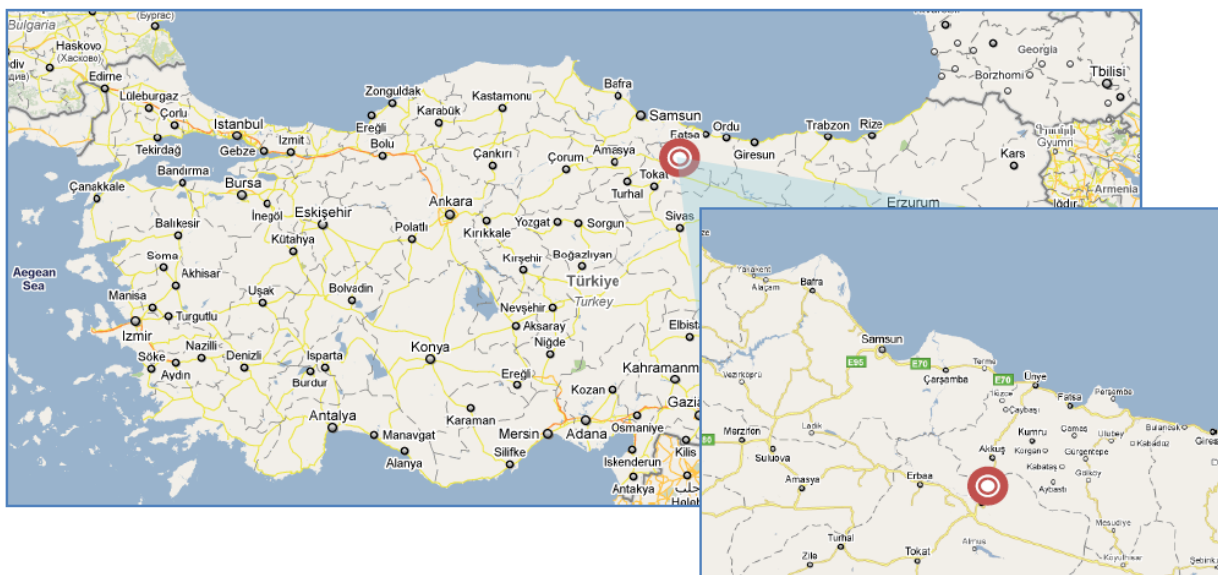


Figure 1. Project location

1.10 Conditions Prior to Project Initiation

Roughly 8,768 people inhabit the Akkus town and 49,799 live in the district. The local economy depends on farming, forestry and grazing livestock plus some handicrafts including carpet weaving.⁸ The Project area was not used for any relevant purpose prior to the Project activity. Local communities are mainly occupied with agriculture-related activities. The Project area, which is sparsely populated, is located on a remote area which is not being used for any relevant irrigation or feedstock purposes.

The Project owner has no control over the baseline GHG emission sources, which will be substituted by the Project. Therefore, it has not been realized for the purpose of subsequent GHG emission reductions.

1.11 Compliance with Laws, Statutes and Other Regulatory Frameworks

Turkey is officially a signatory of the Kyoto Protocol as of 26.08.2009 but has no national legal binding emission reduction targets.

The EIA regulation does not cover the Project and it has been exempted from the regulation with a written declaration by the Ministry of Environment and Forestry. The Project is in compliance with all required relevant regulations but is not a mandatory Project. There is no breach of any legislation and all required permits are obtained.

The following laws and regulations apply for power generation facilities in the host country;

Table 2. Laws and regulations applicable for electricity generation activities

Legal Framework Concerning Electricity Generation
Electricity Market Balancing and Settlement Regulation
Regulation Concerning Electricity Demand Forecast
Communiqué Regarding Connection to and Use of Transmission and Distribution Systems in

⁸ Ref: 2000 census of population (<http://en.wikipedia.org/wiki/Akku%C5%9F>)

the Electricity Market
Communiqué Regarding Preparation of Retail Contract in The Electricity Market
Communiqué Concerning Principles and Procedures for Selection in the Existence of More Than One Application for Generation of Electricity from the Same Source and on the Same Region
Communiqué Regarding Regulation of Market Management Revenue
Regulation on Principles and Procedures for Granting Guarantee of Origin
Electricity Transmission System Supply Reliability and Quality Regulation
Electricity Market Grid Regulation
Communiqué Regarding The Principles and Procedures of Financial Settlement In The Electricity Market
Electricity Market Import and Export Regulation
Communiqué Regarding Regulatory Accounting Guidelines
Electricity Market Distribution Regulation
Communiqué Regarding the Meters to be used in the Electricity Market
Communiqué Regarding Wind and Solar Measurements
Electricity Market Customer Services Regulation
Electricity Market Eligible Consumer Regulation
Electricity Market License Fees
Electricity Market Tariffs Regulation
Electricity Market Licensing Regulation
Electricity Market Law
Other relevant laws:
Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electricity Energy (Nr. 5346)
Energy Efficiency Law (Nr. 5627)
Forest Law (Nr. 6831)

Table 3. Laws and regulations applicable in the host country concerning environmental integrity

Legal Framework Concerning the Environment	
The Environment Law	Packaging Waste Control Regulation
Air Quality Control Regulation	Soil Protection Law
Noise Control Regulation	Air Pollution from Heating Regulation
Water Pollution Control Regulation	Hazardous Chemicals Regulation
Control of Solid Waste Regulation	Waste Batteries Regulation
Environmental Impact Assessment Regulation	Soil Pollution Control Regulation
Medical Waste Regulation	Air Pollution from Vehicles Regulation
Toxic Chemical Substance and Prod Regulation	Medical Waste Control Regulation
Hazardous Waste Regulation	Water Resources Protection from Hazardous Materials Regulation
Waste Oils Control Regulation	Urban Waste Water Treatment Regulation
Industrial Air Pollution Control Regulation	Drinking Water Quality Regulation

1.12 Ownership and Other Programs

1.12.1 Proof of Title

The Project is owned by the Project owner. The generation license issued by the EMRA (Energy Market Regulatory Authority) names the Project owner as the holder of title for this Project activity.

1.12.2 Emissions Trading Programs and Other Binding Limits

The host country, Turkey, has no national emission reduction targets or obligations, as it is not an Annex-B country under the Kyoto Protocol. Therefore there exist no binding limits that would cause the Project's emission reductions be used for other compliance purposes. The Project does not participate in any kind of emissions trading programs.

1.12.3 Participation under Other GHG Programs

The Project is a pure voluntary project and does not seek registration under any other GHG program, because the host country is not eligible for hosting CDM or JI projects.

1.12.4 Other Forms of Environmental Credit

The Project has not created any form of environmental credit and the emission reductions of the Project have not been registered under any other programme. The Project does not intend to pursue any other form of GHG-related credits.

1.12.5 Projects Rejected by Other GHG Programs

The Project has not been rejected by any other GHG programs or standards.

1.13 Additional Information Relevant to the Project

Eligibility Criteria

The Project is not a grouped project and not a debundled part of a larger project.

The Project also complies with the following eligibility criteria;

- VCS v.3 has taken effect on 08.03.2011⁹, so the Project is eligible to use this version.
- VCS v.3 requires project crediting period start date earliest to be 01.01.2002¹⁰. Furthermore, the Project must also complete the validation within two years of the Project Start Date. Both conditions are satisfied for the Project activity.
- VCS v.3 accepts all six Kyoto gases¹¹. The Project aims to reduce CO₂, which is one of these six Kyoto gases.
- VCS v.3 accepts all technologies supported by an approved VCS program. The Project is a hydropower plant, the technology of which is included in the UNFCCC sectoral scope 1.
- VCS v.3 requires the use of one of the VCS program approved project methodologies. The Project activity uses the tools and methodologies of CDM, which is a VCS approved program.
- There is no ODA finance for this Project activity.

⁹ Source: VCS Web Page (<http://www.v-c-s.org/faqs/i-am-developing-project-should-i-use-vcs-version-3-or-20071>)

¹⁰ Source: VCS Standard, Section 3.9 (<http://www.v-c-s.org/sites/v-c-s.org/files/VCS%20Standard%2C%20v3.0.pdf>)

¹¹ Source: VCS Standard, Section 2.1 (<http://www.v-c-s.org/sites/v-c-s.org/files/VCS%20Standard%2C%20v3.0.pdf>)

Leakage Management

Leakage is neglected according to the UNFCCC tools and methodologies that are used. Please refer to Section 2.3 for details.

Commercially Sensitive Information

The public version of the project description does not exclude any items. During validation of the project description, some confidential documents are provided to the DOE that have not been disclosed in detail in this project description, such as;

- The agreement between the carbon consultant and the project owner,
- Expert reports prepared by 3rd parties.

Further Information

N/A

2 APPLICATION OF METHODOLOGY

2.1 Title and Reference of Methodology

The following CDM tools and methodologies are used;

Table 4. Adopted UNFCCC tools and methodologies

Subject	Methodology
Baseline Monitoring	The UNFCCC methodology ACM0002 “Consolidated methodology for grid-connected electricity generation from renewable sources v.12.1.0.
	The UNFCCC methodology “Tool to calculate the emission factor for an electricity system”, v.02.1.0. (EB 60)
Additionality	Tool: The CDM methodological additionality tool, “Tool for the demonstration and assessment of additionality”, v05.2.

2.2 Applicability of Methodology

The selected tools and methodologies are applicable for the Project activity for the following reasons;

Table 5. Assessment of applicability of the adopted methodologies

Subject	Methodology	Justification
Baseline & Monitoring	ACM0002, v.12.1.0.	VCS 2007.1 requires the use of UNFCCC methodologies for baseline and monitoring. The Project does not involve co-generation. The Project is a hydropower plant with renewable components only, with a capacity of more than 15 MW_e. The Project is run-of-river type hydro power plant, which utilizes the potential energy of the flowing water. Therefore, it is a renewable source. The Project is a greenfield investment and does not involve any retrofit, expansion, renewal etc. Its total power output is more than 15 MW. The power density is higher than 4 W/m². The tool version is the most recent version as of the PD preparation date.
	“Tool to calculate the emission factor for an electricity system”, v.02.1.0.	This tool has been used to ensure conservative calculation of the baseline emissions and emission reductions. The tool version is the most recent version as of the PD preparation date. There are no VCS Program Specific Methodologies available for this project type¹².

¹² Source: VCS Web Page (<http://www.v-c-s.org/methodologies/find>)

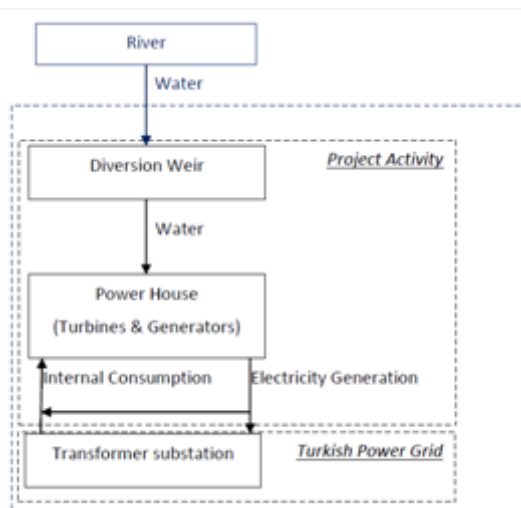
Additionality	“Tool for the demonstration and assessment of additionality”, v05.2.	ACM0002 v.12.1.0 refers to this tool. Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory. The tool version is the most recent version as of the PD preparation date.
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2.3 Project Boundary

The spatial extent of the project boundary includes the Project power plant and all power plants connected physically to the electricity system that the Project power plant is connected to.

The Project is a renewable energy project, aiming at reducing GHG emissions resulting from energy generation activities. Therefore, the Project does not involve any GHG sinks or reservoirs.

For the identification of the baseline scenario, only electricity generation is considered, as the Project has the sole purpose of generating energy. The sources of baseline GHG emissions are defined as the thermal power plants in Turkey, which consume a variety of fuels such as naphtha, coal, lignite or natural gas.



The greenhouse gases and emission sources included in or excluded from the Project boundary are the following;

Table 6. Greenhouse gas sources and sinks

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1	CO ₂	Yes	Main emission source. The dominant emissions from power plants are in the form of CO ₂ , therefore CO ₂ emissions from fossil fuel fired power plants connected to the grid will be accounted for in baseline calculations.
		CH ₄	No	Minor emission source.
		N ₂ O	No	Minor emission source.
		Other		
Project	Source 1	CO ₂	No	The Project does not generate any form of CO ₂ . In urgent cases, a diesel powered generator will be operated for consumption of the personnel and the building, which is negligible.

Source	Gas	Included?	Justification/Explanation
	CH ₄	No	The ACM0002 requires the Project to have a power density higher than 10 W/m ² so that CH ₄ emissions from the reservoir can be neglected. The Project exceeds this threshold: $PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}} = \frac{24.45MW - 0 MW}{113,696 m^2 - 0 m^2} = 215 W/m^2$
	N ₂ O	No	Minor emission source.
	Other		

2.4 Baseline Scenario

The project activity uses the approach described in the ACM0002 v.12.1.0 which describes three alternatives for the baseline. Accordingly, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

The emission factor can be calculated in a transparent and conservative manner as;

- (a) A combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the "Tool to calculate the emission factor for an electricity system"

or

- (b) The weighted average emissions (in t CO₂/MWh) of the current generation mix. The data of the year in which project generation occurs must be used.

The project adopts method (a) to calculate the emission factor. The basic parameters used for calculating baseline emissions of the Project are listed below;

Table 7. Basic parameters used for calculating baseline emissions

Parameter	Value	Data Source / Explanation ¹³
Operation Margin (OM) emission factor (EF _{Grid,OM,y} as tCO _{2e} /MWh)	0.653	Turkey national GHG Inventory TEIAS power generation reports
Build Margin (BM) emission factor (EF _{Grid,BM,y} as tCO _{2e} /MWh)	0.450	TEIAS capacity projection reports IPCC fuel emission values
Combined Margin (CM) emission factor (EF _{Grid,CM,y} as tCO _{2e} /MWh)	0.551	Calculation
Net annual electricity supplied to the grid (MWh)	68,740	Feasibility Study

¹³ Details are provided in Section 4.1.

2.5 Additionality

The following steps are used to demonstrate the additionality of the Project activity according to the “Tool for the demonstration and assessment of additionality”, v.5.2.

STEP 1. Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity

ACM0002 v.12.1.0 requires that the following baseline scenarios should be considered as options:

P1: The project activity not implemented as a CDM project:

This scenario is the installation of the Project activity and its connection to the Turkish national grid without considering carbon revenues.

P2: The continuation of the current situation, i.e. to use all power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance. The additional power generated under the project would be generated in existing and new grid-connected power plants in the electricity system:

This scenario is the continuation of the current situation, where the thermal power plant dominated power generation in Turkey is continued with usual maintenance and addition of new power plants to the grid with historical trends.

Outcome of sub-step 1a: Three scenarios are identified which are realistic and credible alternatives available to the Project owner.

Sub step 1b: Consistency with mandatory laws and regulations

All the alternatives listed above are in compliance with the regulatory requirements of the public authorities in Turkey and neither of these is prohibited from prevailing rules and regulations. Please refer to **Table 2** and **Table 3** for details. There are no legal constraints that would prevent the Project owner to implement any of the alternatives listed above.

Step 2. Investment analysis

The Project activity will export the electricity to the Turkish grid and will reduce a proportional amount of GHG emissions that would otherwise have been generated by fossil fuel power plants in the grid, resulting in the reduction of carbon intensity in the grid. The electricity delivered to the grid and the grid emission factor becomes the basis in the calculation of emissions reductions. Estimated annual emission reductions resulting from the Project activity is approximately 37,883 tCO_{2e}/a.

This step involves the determination whether the proposed Project activity is not:

- a) The most economically or financially attractive; or
- b) Economically or financially feasible, without the revenue from the sale of emission reductions.

Sub-step 2a: Determine appropriate analysis method

The following options are available to conduct the benchmark analysis;

Table 8. Selection of investment analysis option

	Investment Analysis Options	Explanation
Option (I)	Simple cost analysis	Not applicable to the project because the project activity generates the revenue from the sales of electricity generation.
Option (II)	Investment comparison analysis	Not applicable because an investment comparison analysis shall be used if the proposed baseline scenario leaves the project participant no other choice than to make an investment to supply the same (or substitute) products or services.
Option (III)	Benchmark analysis	This option (III) is chosen.

ACM0002 provides two options for the investment analysis. The benchmark analysis option is selected because more than one alternative is remaining after Step 2 and the remaining alternatives include scenarios P1 and P2.

Sub-step 2b: Option III. Apply benchmark analysis

The likelihood of the development of this Project will be determined by analyzing its “Project IRR” based on the scenarios with and without the sale of carbon credits. The chosen benchmark for the Project is the (pre-tax) WACC, as the CDM “Guidelines on the Assessment of Investment Analysis”, v.3 (referred to as “Guidelines” hereafter) suggests.

Sub-step 2c: Calculation and comparison of financial indicators

The following financial figures are used for calculations;

Table 9. Parameters used for benchmark analysis

Parameter	Figure	Source / Justification
Discount rate (WACC)	10.1%	Calculated. The Guidelines suggest WACC as an appropriate benchmark.
Electricity Price/kWh	5.5 EUR cents	Renewable Energy Law
Capital Expenditure	24,078 TEUR	Feasibility Study ¹⁴

¹⁴ Adjusted for capacity increase (for electromechanical equipment)

O&M Costs	602 TEUR	Engineering Study
Investment horizon	20 years	The Guidelines suggest a minimum of 10 years and a maximum of 20 years.
Corporate tax rate	20%	This is the official corporate tax rate in the host country (www.invest.gov.tr)
Equity/Debt ratio	25/75	Assumption at the time of investment decision.
Loan Interest Rate	7.25 %	Assumption at the time of investment decision.
Power Generation	68,740 MWh	Feasibility Study

The WACC is calculated with the following formulae;

$$WACC_{posttax} = \frac{E}{E+D} * CoE + \frac{D}{E+D} * CoD * (1 - T_c) \quad [1]^{15}$$

$$WACC_{pretax} = \frac{WACC_{posttax}}{(1-T_c)} \quad [2]^{16}$$

Financial expenditures are not considered in the calculations. The explanations of the parameters are provided below;

Table 10. Definitions of parameters

Parameter	Explanation
$WACC_{posttax}$	Weighted average cost of capital after tax
$WACC_{pre-tax}$	Weighted average cost of capital before tax
E	Equity ratio in the total investment volume
D	Debt ratio in the total investment volume
CoE	Cost of equity
CoD	Cost of debt
T_c	Corporate tax rate

Further details on the calculation of WACC are provided to the validating DOE. The Project IRR calculations are carried out for both scenarios with and without considering carbon credits. The calculations conservatively assume that the revenues from electricity sales are obtained without delay, that there are no major unforeseen malfunctions and that water flow levels will be as expected. The results are provided below;

Table 11. Comparative table of the financial indicators with and without carbon revenues

Scenario	Project IRR
Without carbon revenues	8.8%
Benchmark discount rate	10.1%
With carbon revenues	10.1%

¹⁵ Source: <http://www.investopedia.com/terms/w/wacc.asp>

¹⁶ Source: <http://www.zanders.nl/publicaties/documents/WACCpart8.pdf>

The results indicate that without carbon credits, the financial performance of the Project is not feasible. Under these circumstances, the Project would not be attractive for the investor.

Sub-step 2d: Sensitivity analysis

The purpose of the sensitivity analysis is to examine how the uncertainty of a single variable such as total investment cost and net power output will have an impact on the Project IRR. The assumptions set in the analysis process are increasing/decreasing $\pm 15\%$ from the original level of the parameters. The following parameters are identified which would have a significant impact on the financial results;

- Revenue from electricity sales
- Capital expenditure
- Operation and maintenance costs

The results are shown in the figure below;

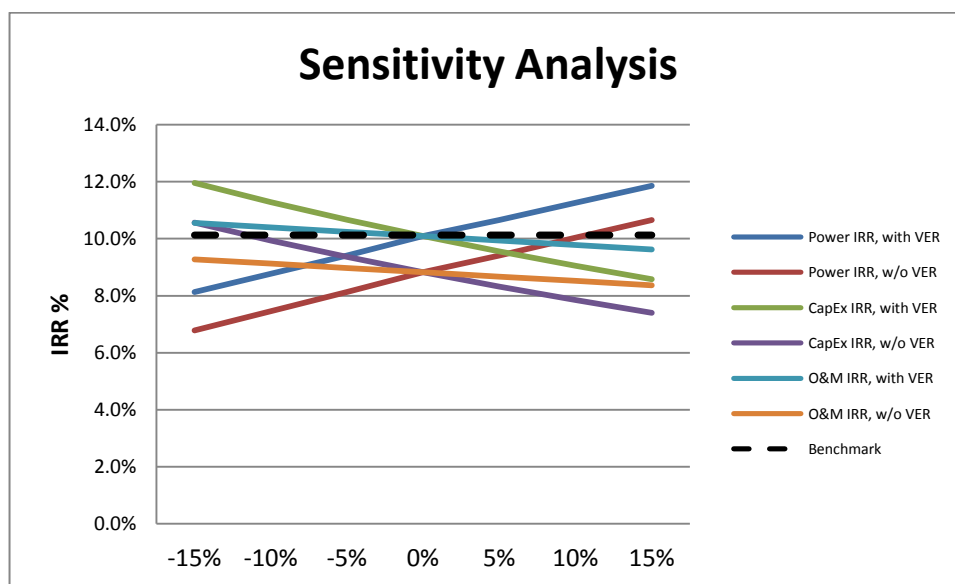


Figure 2. Results of the sensitivity analysis

Here it can be seen that revenues from power sales and CapEx have considerable impact on the Project IRR, whereas O&M costs have limited impact. The results indicate the following:

- In case the electricity price decreases by 5%, the Project becomes unfeasible, even with carbon credits. This is highly unlikely, since the electricity price is guaranteed by law. If revenues from the sales of electricity would increase by more than 10%, the Project would have become feasible without the use of carbon credits. This is not very likely, since the technical design is based on long historical flow rates and the renewable energy law has recently been revised, with no improvement in electricity price granted to hydropower plants. Considering potential cost of foreign exchange risks¹⁷, additional transmission and connection fees and deductions made by TEIAS, and historical electricity price levels any significant upside variation of electricity price can be seen as a fairly low probability.

¹⁷ The guaranteed price of electricity set out by the law for renewable energy is given in EUR cents and not local currency (Turkish Lira), the exchange rate risk of which can be covered by purchasing hedging services.

- b) In case the capital expenditures increase by 5%, the Project becomes unfeasible, even with carbon credits. This is likely, since construction costs can change based on material costs and unforeseen technical difficulties. If capital expenditures decrease by 15%, then the Project becomes feasible without the use of carbon credits. However this is most unlikely, because the cost planning of the Project owner so far indicates that their projects exceed their budgets. Considering that the cash flow of the Project includes a fair residual value at the end of investment horizon, and that the Project owner has assigned its mother company, Atac Insaat, for construction works, it is less likely to expect the CapEx to drop significantly.
- c) The operating and maintenance expenses do not have a significant impact on the Project profitability and even an increase or decrease by 15% does not result in exceeding the benchmark. A higher change in O&M expenses are not realistic, as extreme cost minimisation efforts could compromise proper equipment maintenance and could lead to poor working conditions or underperformance.

Within the reasonable variation of the total investment, revenues from electricity sales and O&M costs, the calculated IRRs are below the benchmark IRR, thus the project keeps economically unfeasible without carbon revenues. So it is demonstrated that the proposed Project activity is not the most economically or financially attractive and not financially feasible without the revenue from the sale of carbon credits.

Outcome of Step 2: The proposed CDM project activity is unlikely to be the most financially/economically attractive.

Step 3. Barrier analysis

This step is not applied.

Step 4. Common practice analysis

Sub-step 4a: Analyze other activities similar to the proposed project activity

This step involves providing an analysis of any other activities that are operational and that are similar to the proposed project activity. Projects are considered similar if they are in the same country/region and/or rely on a broadly similar technology, are of a similar scale, and take place in a comparable environment with respect to regulatory framework, investment climate, access to technology, access to financing, etc. Other CDM project activities are not to be included in this analysis.

Under this guidance, the similar project activities are defined as the following:

- All power plants that have a generation license from EMRA are considered, as generation license means the first major step concerning the intent to invest in the host country.
- Similar technology and regulatory framework: Only hydropower plants that are IPPs are considered. Publicly owned power plants or Autoproducers are excluded.
- Same region: Only hydropower plants in the same DSI region¹⁸ are considered. DSI categorizes Turkey in certain hydrological regions based on their geographical, precipitation, geological and managerial criteria.
- Technical similarities: Power plants that have a significantly larger or smaller installed capacity are excluded, along with power plants that have a significantly higher plant load factor (i.e. ratio of energy generation to installed capacity).

¹⁸ Source: <http://www2.dsi.gov.tr/bolge/bolgeler.htm>

Also according to the guidance provided in the “Tool for the demonstration and assessment of additionality”, v.5.2, carbon projects are excluded.

Sub-step 4b: Discuss any similar Options that are occurring

Based on the approach explained above, the following project have been identified to be similar;

Table 12. Identified similar project activities¹⁹

Company name	Province / Project name	Capacity, MW _e	Capacity in operation, MW _e
GA Elektrik Enerjisi Üretim Satış Sanayi Ve Ticaret Limited Şirketi	Amasya İli, Umutlu HES	22.28	
Masat Enerji Elektrik Üretim ve Tic.Ltd.Şti..	Amasya	22.5	22.5
Temmuz Elektrik Üretim Ltd.Şti.	Ordu/MURAT HES	32.45	
Temmuz Elektrik Üretim Ltd.Şti.	Tokat İli, Onur Regülatörü ve HES	18.6	
Turhal Enerji Üretim Elektrik ve Tic. Ltd.	Amasya İli Osmançık HES	12.46	

According to the “Tool for the demonstration and assessment of additionality”, v.5.2, any other activities that are operational shall be considered. The **Table 12** shows that only one other project is in operation in Amasya with 22.5 MW capacity (Tasova HES). Other projects shall not be considered as similar. The Project activity will make use of four diversion weirs, instead of one, showing the technical differences with Tasova HES. This analysis clearly illustrates that the implementation of Project activity is not common practice.

Prior Consideration of the CDM

The definition of "starting date" is set out in the CDM Glossary of Terms as the earliest date at which either the implementation or construction or real action of a project activity begins. The construction of the Project activity has started on 01.10.2010. The equipment order has been signed on 05.06.2008. There is no loan agreement as of the date of PDD preparation. The earliest of these dates is the equipment order date. The Project owner has signed a carbon asset development agreement with the carbon consultants before this date, demonstrating prior consideration of carbon credits. The chronological history of the Project is provided below;

Table 13. Timeline of the Project

Date	Development
30.01.2007	Project Owner Nisan Enerji is purchased by the investor.
04.03.2008	Water use agreement date with DSI.
24.04.2008	An ERPA proposal is sent by an international carbon consultant to the Project Owner (first consideration of carbon revenues).
01.05.2008	The electricity generation license is issued by EMRA.

¹⁹ Source: EMRA generation licenses as of 01.03.2011 (<http://www2.epdk.org.tr/lisanss/elektrik/lisansdatabase/verilentesistipi.asp>)

June 2008	The technical due diligence report is prepared by the lender's engineer.
05.06.2008	Equipment order is signed.
06.10.2008	Project Owner signs a board decision to include carbon revenues in the Project.
February 2009	The technical feasibility report is prepared.
09.07.2009	Project Owner signs a carbon asset development service agreement with Mavi Consultants
12.12.2009	The document stating that an "EIA is not required" is obtained.
20.07.2010	Some of the shares of the company are sold to Boydak Enerji.
01.10.2010	Project mobilisation date.
01.10.2012	Expected Project operation start date.

The Project is completely a voluntary project that is established for the sole purpose of generating electricity. Its realization and operation are not mandated by any enforced law, statute or other regulatory framework and is therefore a regulatory surplus.

The financially more viable alternative to the Project activity is that the electricity generated by the Project activity would have otherwise been provided by the national grid, and the emission reductions of the Project would not have been realised in the absence of carbon credits.

Outcome of Step 4: Similar activities are observed, but essential distinctions between the project activity and similar activities can reasonably be explained. Therefore sub-steps 4a and 4b are satisfied and the proposed Project activity is additional.

2.6 Methodology Deviations

N/A

3 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

3.1 Baseline Emissions

Emissions reductions achieved by the Project activity are considered equivalent to the emissions avoided in the baseline scenario by displacing the grid electricity. The baseline emissions (BE_y) include only CO₂ emissions that are displaced due to the Project activity and the approach adopted for calculating emission reductions due to implementation of Project activity is according to the “Tool to calculate the emission factor for an electricity system”, v.02.1.0 (EB 60). The “Tool to calculate the emission factor an electricity system” is applied in the context of the Project in the following seven steps;

Step 1. Identify the relevant electric power system

For determining electricity emission factors, a project electricity system is defined by the spatial extent of power plants that are physically connected through transmission and distribution lines to the project activity and that can be dispatched without significant transmission constraints.

The Project will be connected to the Turkish electricity grid, which is an interconnected grid to which all power plants are connected to. Therefore, the Project electricity system is identified as the national grid. There are no regional grids in the host country. Turkey imports electrical power from and exports to neighbour countries, which are defined as connected electricity systems for the Project. According to the baseline methodology, imports are regarded as a power source delivering electricity to the grid with an OM emission factor of 0 tCO₂/MWh, since some small amount of electricity being imported is purchased from connected electricity systems located in other countries.

Step 2. Choose whether to include off-grid power plants in the project electricity system

The “Tool to calculate the emission factor for an electricity system” offers two options to calculate the operating margin and build margin emission factor:

- a) Option I: Only grid power plants are included in the calculation.
- b) Option II: Both grid power plants and off-grid power plants are included in the calculation.

Option I is applied and off-grid power plants are excluded from the calculations.

Step 3. Select a method to determine the operating margin (OM)

The baseline methodology allows a choice among four methods for the calculation of OM emission factor;

- a) Simple OM, or
- b) Simple adjusted OM, or
- c) Dispatch Data Analysis OM, or
- d) Average OM

There exist no publicly available data for the dispatch data analysis (c) or for the simple adjusted OM (b). There exists no nuclear power plant in Turkey, and there is no indication that coal or lignite is obviously used as must-run. Hydro, geothermal, wind power plants and other renewables are included as low-cost/must-run resources (hereafter referred as “lc-mr”), which

turns out to be 21% of the total electricity generation on average in the last 5 years, as shown in Table 14. The share of lc-mr resources is less than 50% and available data are limited, therefore simple OM (a) has been selected. The electricity generation breakdown based on fuel sources and their shares are provided below;

Table 14. Gross and Net Electricity Generation [GWh] in Turkey²⁰

Fuel Type	2005	2006	2007	2008	2009	Avg. Share
Coal	13,246	14,217	15,136	15,858	16,596	8%
Lignite	29,946	32,433	38,295	41,858	39,090	20%
Fuel Oil	5,121	4,232	6,470	7,209	4,440	3%
Diesel Oil	3	58	13	266	346	0%
LPG	34	0	0	0	0	0%
Naphtha	326	50	44	44	18	0%
Natural Gas	73,445	80,691	95,025	98,685	96,095	48%
Renewables	122	154	214	220	340	0%
Hydro	39,561	44,244	35,851	33,270	35,958	20%
Geothermal & Wind	153	221	511	1,009	1,931	0%
Gross Gen. Total [GWh]	161,956	176,300	191,558	198,418	194,813	100%
Net Generation	155,469	169,543	183,340	189,762	186,619	
Net Delivery Ratio	96%	96%	96%	96%	96%	
Imports	636	573	864	789	812	
Gross Total excl. LC-MR	122,120	131,681	154,983	163,919	156,583	
Net Total excl. LC-MR	117,864	127,208	149,198	157,558	150,810	

Accordingly, the emission factor can be calculated using either of the two data vintages:

- Ex-ante option, where a 3-year generation-weighted average based on the most recent data is used. Monitoring and recalculation of the emission factor is not required, or,
- Ex-post option, where the data of the year is used, in which the project activity displaces grid electricity. Yearly update of the emission factor is required.

The ex-ante option (a) is selected to carry out the baseline methodology for the Project. This means that the OM will be calculated as a weighted average of the latest 3 years and will be used as a constant throughout the crediting period.

Step 4. Calculate OM emission factor according to the selected method

The “Tool to calculate the emission factor for an electricity system” offers two options to calculate $EF_{grid,OMsimple,y}$:

- Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit, or,
- Option B: Based on data on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Detailed data on the individual power plants connected to the national electricity grid necessary for applying option A is not available. Therefore Option A cannot be used. Since only renewable power generation are considered as low-cost / must-run power sources and since the quantity of electricity supplied to the grid by these individual sources is known, option B is applicable and used to calculate the Simple OM emission factor. $EF_{grid,OMsimple,y}$ using option B is calculated

²⁰ Source: TEIAS ([http://www.teias.gov.tr/istatistik2009/32\(75-09\).xls](http://www.teias.gov.tr/istatistik2009/32(75-09).xls), [http://www.teias.gov.tr/istatistik2009/30\(84-09\).xls](http://www.teias.gov.tr/istatistik2009/30(84-09).xls))

based on:

- The net electricity supplied to the grid by all power plants serving the system, not including lc-mr power plants / units, and,
- The total emissions (ktCO_{2e}) resulting from power generation in the host country, submitted as in the latest GHG inventory of Turkey to UNFCCC.

2007-2009 national emissions figure stemming from electricity generation is publicly available and used for OM calculations, as shown below;

Table 15. Calculation of the OM emission factor

Operating Margin Emission Factor	2007	2008	2009
Official CO ₂ Emissions ²¹ [ktCO ₂]	100,662	101,473	96,286
Net electricity delivered, excl. lc-mr [GWh]	149,198	157,558	150,810
Emission Factor [tCO ₂ /MWh]	0.675	0.644	0.638
Weight of generation, %	32.8%	33.9%	33.3%
Operating Margin [tCO ₂ /MWh]	0.653		

The national GHG inventory of Turkey is used because it is the officially submitted amount of emissions resulting from power generation in Turkey. It is calculated based on expert opinions, detailed fuel and power plant information and based on UNFCCC reporting requirements. The inventory does not particularly list the power plants it includes for the emission base. As the publicly available information about fuel types, their quality and combustion technologies adopted by power plants in Turkey is quite limited, using official emissions figures are more dependable and have a higher quality than calculating them with default figures. The UNFCCC tools and methodologies also suggest prioritizing locally available data, if such data are available. Therefore emissions are taken from the official GHG inventory rather than calculations.

Step 5. Identify the cohort of power plants to be included in the build margin

According to the “Tool to calculate the emission factor for an electricity system”, the sample group of power units m used to calculate the build margin consists of either:

- The set of five power units that have been built most recently, or
- The set of power capacity additions in the electricity system that comprise 20% of the system generation (in MWh) and that have been built most recently.

The “Tool to calculate the emission factor for an electricity system” requires the sample list of recent capacity additions to constitute at least 20% of the system generation. For conducting the calculations, option (b) is selected, because this option results in a larger electricity generation.

The data of the most recent commissioned power plants are being published by the Turkish Electricity Transmission Company (TEIAS) on an annual basis. In order to meet the 20% threshold, recently built power plants between 2005-2009 are included in the cohort. The threshold requirement is satisfied;

²¹ Source: National GHG inventory report of Turkey, as submitted to UNFCCC
(http://unfccc.int/files/national_reports/annex_i_ghg_inventories/national_inventories_submissions/application/zip/tur-2011-crf-13apr.zip)

Table 16. Generation of Recent Capacity Additions²²

Selection of the Sample Power Plant List	
UNFCCC Threshold	20%
System Generation ²³ [GWh]	194,813
Minimum Threshold (20%)	38,962.6
Total of selected power plants [GWh]	39,884.3

In this selection, those power plants are not included that are capacity additions or reductions, performance revisions, retrofits, isolated plants, dismantlings or CDM project activities. The final list of the cohort of power plants is available in the Annex for further details.

The “Tool to calculate the emission factor for an electricity system” offers the choice between two data vintages to calculate the BM:

- Option 1. For the first crediting period, BM emission factor is calculated ex-ante based on the most recent information available on units already built for sample group m at the time of PDD submission to the DOE for validation, or,
- Option 2. For the first crediting period, the BM emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity.

Option 1 is selected. This data vintage remains fixed during the first crediting period and will be updated for the second crediting period.

No recent or future addition to the national transmission capacity is reported or foreseen, which would enable Turkey to significantly increase its electricity imports. Therefore, a likely transmission line construction is not considered in BM emission factor calculations.

Step 6. Calculate the build margin emission factor

For the power units in Turkey, only data on electricity generation and the fuel types used is available. Therefore, Option A2 (as in Step 4, Simple OM approach) is used and the emission factor is determined based on the CO₂ emission factor of the fuel type used and the efficiency of the power unit.

The build margin emission factor is the generation-weighted average emission factor (tCO₂/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows;

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} * EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad [3]$$

Table 17. Explanations of parameters used for calculations

Parameter	Explanation
EF _{grid, BM,y}	Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
EG _{m,y}	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)

²² Source: TEIAS annual capacity projection reports (<http://www.teias.gov.tr/istat2004/7.xls>, <http://www.teias.gov.tr/istatistik2005/7.xls>, www.teias.gov.tr/projeksiyon/KAPASITE%20PROJEKSIYONU%202007.pdf, <http://www.teias.gov.tr/projeksiyon/KAPASITEPROJEKSIYONU2008.pdf>, <http://www.teias.gov.tr/projeksiyon/KAPASITEPROJEKSIYONU2009.pdf>, <http://www.teias.gov.tr/projeksiyon/KAPASITE%20PROJEKSIYONU%202010.pdf>)

²³ Source: TEIAS ([http://www.teias.gov.tr/istatistik2009/32\(75-09\).xls](http://www.teias.gov.tr/istatistik2009/32(75-09).xls))

$EF_{EL,m,y}$	CO ₂ emission factor of power unit m in year y (tCO ₂ /MWh)
m	Power units included in the build margin
y	Most recent historical year for which power generation data is available

Calculations for the BM emission factor $EF_{grid,BM,y}$ is based on the most recent information available on the plants already built for sample group m at the time of PDD submission. Using the sample list of recently built power plants, the Build Margin Emission Factor, $EF_{grid,BM}$ is found as;

Table 18. Calculation of the BM emission factor

Capacity Additions	Total Generation [GWh]	Indicative Efficiency ²⁴	Emission Intensity [tCO ₂ /TJ] ²⁵	Emissions [ktCO ₂]	Build Margin [tCO ₂ /MWh]
Coal	1125.0	40% ²⁶	89.5	906.2	
Lignite	11455.8	40% ²⁶	90.9	9371.9	
Fuel Oil	2513.6	46%	75.5	1485.2	
Diesel Oil	0.0	46%	72.6	0.0	
LPG	0.0	60%	61.6	0.0	
Naphtha	0.0	46%	69.3	0.0	
Natural Gas	18938.2	60%	54.3	6170.1	
Renewables	139.1		0	0.0	
Hydro	4962.3		0	0.0	
Geothermal & Wind	750.4		0	0.0	
Total	39,884.3			17,933.4	0.450

Power plant-specific data are unavailable in Turkey; therefore CO₂ emissions are calculated based on fuel type consumed in sample power plants/units.

Step 7. Calculate the combined margin emissions factor

The combined margin (CM) emissions factor ($EF_{grid,CM,y}$) is calculated as a weighted average of Operating Margin emission factor and Build Margin emission factor as follows:

$$EF_{grid,CM,y} = W_{OM} * EF_{grid,OM,y} + W_{BM} * EF_{grid,BM,y} \quad [4]$$

Table 19. Explanations of parameters used for calculations

Parameter	Explanation
$EF_{grid,CM,y}$	Combined margin CO ₂ emissions factor in year y (tCO ₂ /MWh)
$EF_{grid,BM,y}$	Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)

²⁴ Source: UNFCCC "Tool to calculate emission factor for an electricity system", Annex 1

²⁵ Source: Lower values in the IPCC 2006 Guidelines V2, Ch1, Table 1.4

²⁶ The average design efficiency of coal power plants in Turkey are 38%. Therefore assuming CFBS technology with 40% efficiency is appropriate and conservative. (Source: http://www.esmap.org/esmap/sites/esmap.org/files/Rpt_TurkeyESM273.pdf)

$EF_{grid,OM,y}$	Operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
W_{OM}	Weight of operating margin emissions factor (0.5 by default)
W_{BM}	Weight of build margin emissions factor (0.5 by default)

$$EF_{grid,CM,y} = 0.5 * 0.653 \text{ tCO}_2/\text{MWh} + 0.5 * 0.450 \text{ tCO}_2/\text{MWh} = 0.551 \text{ tCO}_2/\text{MWh}$$

The baseline emissions (BE_y) are obtained as:

$$BE_y = EG_{BL,y} * EF_{CO_2,grid,y} \quad [5]$$

Where:

$$EG_{BL,y} = EG_{facility,y}$$

$$EF_{CO_2,grid,y} = EF_{grid,CM,y}$$

Table 20. Explanations of parameters used for calculations

Parameter	Explanation
BE_y	Baseline emissions in year y (tCO ₂)
$EG_{BL,y}$	Quantity of net electricity supplied to grid by the Project activity in year y (MWh)
$EF_{CO_2,grid,y}$	CO ₂ emission factor of the grid (tCO ₂ /MWh)
$EG_{facility,y}$	Quantity of net electricity supplied to grid by the Project plant in year y (MWh/a)
$EF_{grid,CM,y}$	Combined margin CO ₂ emissions factor in year y (tCO ₂ /MWh)

$$BE_y = 68,740 \text{ MWh} * 0.551 \text{ tCO}_2/\text{MWh} = \mathbf{37,883 \text{ tCO}_2}$$

3.2 Project Emissions

Project emissions are taken as zero, in agreement with the used tool and methodology. The Project has negligible GHG emissions, because;

- The Project Activity only involves generation of electricity from renewable sources,
- Methane emissions from the reservoir is negligible (Project's power density is above 10 W/m²),
- No anthropogenic GHGs by sources outside the project boundary that are significant, measurable and attributable to the project activity could be identified.

Please see Table 6 for further information.

3.3 Leakage

ACM0002 requires that leakage should be considered if the energy generating equipment is transferred from another activity. This is not the case, therefore leakage is taken as zero.

3.4 Summary of GHG Emission Reductions and Removals

The emissions reductions are calculated as the following:

$$ER_y = BE_y - PE_y - L_y \quad [6]$$

Table 21. Explanations of parameters used for calculations

Parameter	Explanation
ER_y	Emission reductions in year y (tCO ₂)
BE_y	Baseline emissions in year y (tCO ₂)
PE_y	Project emissions in year y (tCO ₂)
L_y	Leakage emissions in year y (tCO ₂)

$$PE_y = L_y = 0$$

Therefore, $ER_y = BE_y = 37,883 \text{ tCO}_2$

The estimated baseline emissions and emissions reductions are provided below;

Table 22. Summary of ex-ante estimation of emission reduction

Years	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2012	9,860	0	0	9,860
2013	37,883	0	0	37,883
2014	37,883	0	0	37,883
2015	37,883	0	0	37,883
2016	37,883	0	0	37,883
2017	37,883	0	0	37,883
2018	37,883	0	0	37,883
2019	37,883	0	0	37,883
2020	37,883	0	0	37,883
2021	37,883	0	0	37,883
2022	28,023	0	0	28,023
Total	378,831	0	0	378,831

4 MONITORING

4.1 Data and Parameters Available at Validation

Data Unit / Parameter:	Official emissions from power generation
Data unit:	tCO ₂
Description:	National GHG Inventory of Turkey submitted to UNFCCC
Source of data:	UNFCCC web site ²⁷
Value applied:	Used in: Table 15 .
Justification of choice of data or description of measurement methods and procedures applied:	These are 2007-2009 official emissions data submitted to UNFCCC for power generation facilities in Turkey.
Any comment:	

Data Unit / Parameter:	GEN _{j,y}
Data unit:	MWh
Description:	The gross electricity generation by fuel type j in year y (2005-2009) by the system
Source of data:	TEIAS web site ²⁸
Value applied:	Used in: Table 14 The sum of GEN _{j,y} for each fuel type in year y gives the System Gross Electricity Generation _y .
Justification of choice of data or description of measurement methods and procedures applied:	TEIAS annually publishes official data regarding electricity generation and is the only source that provides information with most details.
Any comment:	

Data Unit / Parameter:	System Net Electricity Generation
Data unit:	MWh
Description:	The difference between the total quantity of electricity generated by power plants/units and

²⁷ Source: Worksheet 1s1 in the http://unfccc.int/files/national_reports/annex_i_ghg_inventories/national_inventories_submissions/application/zip/tur-2011-crf-13apr.zip

²⁸ Source: [http://www.teias.gov.tr/istatistik2009/32\(75-09\).xls](http://www.teias.gov.tr/istatistik2009/32(75-09).xls)

	the auxiliary electricity consumption of power plants/units.
Source of data:	TEIAS web site ²⁹
Value applied:	Used in: Table 14
Justification of choice of data or description of measurement methods and procedures applied:	TEIAS annually publishes official data regarding electricity generation and is the only source that provides information with most details.
Any comment:	

Data Unit / Parameter:	Net Delivery Ratio
Data unit:	%
Description:	The ratio of the System Net Electricity Generation to the System Gross Electricity Generation in year y.
Source of data:	Calculation: By dividing System Net Electricity Generation _y by System Gross Electricity Generation _y
Value applied:	Used in: Table 14
Justification of choice of data or description of measurement methods and procedures applied:	Electricity delivered to the grid by each power plant/unit is unavailable. The Net Delivery Ratio _y is used for approximating the net electricity amount delivered to the grid by power plants/units except lc-mr sources.
Any comment:	This is a conservative assumption, since in general thermal power plants consume more energy for auxiliaries than e.g. hydro plants.

Data Unit / Parameter:	GEN _y
Data unit:	MWh
Description:	The system net electricity delivered in year y by the total of all power plants in the grid, excluding lc-mr sources.
Source of data:	Calculation: 1. Gross electricity generation excluding lc-mr sources = Total gross electricity generation from all sources - total gross electricity generation from lc-mr sources 2. Net electricity generation excluding lc-mr

²⁹ Source: [http://www.teias.gov.tr/istatistik2009/30\(84-09\).xls](http://www.teias.gov.tr/istatistik2009/30(84-09).xls)

	sources = Gross electricity generation except lc-mr sources * Net Delivery Ratio 3. Net electricity delivered to the grid except lc-mr sources = Net electricity generation except lc-mr sources + Imports
Value applied:	Used in: Table 14
Justification of choice of data or description of measurement methods and procedures applied:	Net electricity delivered to the grid by each power plant/unit or by fuel type is unavailable. Therefore this parameter is approximated by calculations.
Any comment:	

Data Unit / Parameter:	EF _{CO₂,i,y}
Data unit:	tCO ₂ /TJ
Description:	CO ₂ emission factor of fossil fuel type i in year y
Source of data:	The lower limits of the 95% confidence interval stated in the “2006 IPCC Guidelines for National Greenhouse Gas Inventories”, Volume 2, Chapter 1 (energy) Table 1.4.
Value applied:	Used in: Table 18
Justification of choice of data or description of measurement methods and procedures applied:	This is in line with the “Tool to calculate the emission factor for an electricity system”.
Any comment:	

Data Unit / Parameter:	Electricity Import _y
Data unit:	GWh
Description:	Net electricity imports delivered to the grid in year y
Source of data:	TEIAS web site ³⁰
Value applied:	Used in: Table 14
Justification of choice of data or description of measurement methods and procedures applied:	
Any comment:	This is the total electricity imported and delivered to the national grid from connected electricity systems from neighbour countries.

³⁰ Source: [http://www.teias.gov.tr/istatistik2009/30\(84-09\).xls](http://www.teias.gov.tr/istatistik2009/30(84-09).xls)

Data Unit / Parameter:	$\eta_{m,y}$
Data unit:	%
Description:	Average net energy conversion efficiency of power unit m in year y
Source of data:	UNFCCC methodological tool "Tool to calculate the emission factor for an electricity system" Annex I
Value applied:	Used in: Table 18
Justification of choice of data or description of measurement methods and procedures applied:	Default efficiency figures represent best available techniques and should be considered to be conservative.
Any comment:	There are no official efficiency values available for recent power plants based on each power plant or each fuel type. Therefore default values are assumed.

4.2 Data and Parameters Monitored

Data Unit / Parameter:	$EG_{\text{facility},y}$
Data unit:	kWh
Description:	Net electricity supplied to the grid by the Project activity in year y
Source of data:	Primary source: PMUM web site Secondary source: invoices and TEIAS protocols
Description of measurement methods and procedures to be applied:	Measurement will be carried out by accurate electricity meters. The collected data will be, once approved by TEIAS, used to calculate the net electricity supplied by the Project.
Frequency of monitoring/recording:	Continuous monitoring and recording. Monthly reporting.
Value applied:	68,740 MWh
Monitoring equipment:	The measurement will be carried out and recorded continuously by a primary and a secondary measurement device. The Project has to comply with the "Communiqué for Metering Devices to be used in the Electricity Market", and has to install IEC-EN 60687 0.5 class measurement instruments that have a minimum accuracy of 0.5% ³¹ , which satisfies UNFCCC requirements.

³¹ Source: EMRA web site (<http://www.epdk.gov.tr/web/elektrik-piyasasi-dairesi/44>)

QA/QC procedures to be applied:	There will be no sampling and all measured data will be used for monitoring. Measurement equipment will be calibrated and their accuracy will be ensured by TEAIS. Both measurement devices will be approved by TEIAS, and the calibration will also be managed by TEIAS. According to the “Communiqué for Metering Devices to be used in the Electricity Market”, Article 9, paragraph b, the periodic calibration shall be made every 10 years³². Both the primary and secondary devices are sealed during first operation of the plant. In case there is a significant inconsistency (assessed by TEIAS) between two devices, the meters will be calibrated by TEIAS before the 10 years period. The Project owner can also request calibration of the meter(s). The calibration stations are inspected by the central organization of the same Ministry³³. Both metering instruments are sealed and only accessible by TEIAS personnel. PMUM data is also of high quality, as the data is used for invoicing purposes and is accessible online by the Project owner, minimizing the risks of data corruption, data archiving, loss of information etc. Generation figures from PMUM reports will be copied-pasted to the spreadsheet to prevent potential typos.
Calculation method:	The net electricity will be calculated by: - Subtracting self consumption³⁴ value from gross generation³⁵ value for each month to find the net electricity supplied to the grid. - Adding up all monthly net electricity values to calculate the total net electricity supplied to the grid during the monitoring period. - Multiplying the total net electricity value with the CM emission factor.
Any comment:	The metering devices do not measure the net electricity generation but the gross generation

³² “Communiqué for Metering Devices to be used in the Electricity Market”, Article 9, paragraph b. (Document already submitted to the DOE.)

³³ Source: “Communiqué for Metering Devices to be used in the Electricity Market”, Article 10

³⁴ “UECM” in PMUM reporting.

³⁵ “ISVM” in PMUM reporting.

	and self consumption separately. Concerning invoicing, TEIAS deducts a certain amount of electricity supplied to the grid by the Project in order to compensate its transmission line losses in its national grid. According to ACM0002, emissions reductions are based on net electricity supplied to the grid, therefore this deduction is not incorporated into the monitoring calculations.
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4.3 Description of the Monitoring Plan

Procedure:

For monitoring, the UNFCCC methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” v.12.1.0 is used. The GHG emission reductions are achieved through electricity generation by the Project. Therefore, the net electricity supplied by the Project, which is the difference of gross electricity generation and Project’s own electricity consumption, will be monitored. For this purpose, continuously collected electricity generation and consumption data will be reported and recorded monthly. Project emissions or leakage during crediting period will not be monitored.

The measured data are the monthly electricity generation and consumption values. These data are read and reported by TEIAS and can be called from the PMUM web site during the following month.

The Project activity itself consumes some amount of energy for operation. This amount is used from the generation of the Project, if the Project is operating. The remaining energy, gross generation is then supplied to the grid. Sometimes the Project will not be operating and taking some electricity from the grid for internal consumption, which is the consumption. Therefore there will be no relevant data that shows internal consumption when the Project is in operation, as it is automatically deducted. The consumption data only shows the amount which is used from the grid. Both gross generation and consumption data are measured and recorded by the same measurement devices.

There will be two measurement devices on site that will continuously monitor electricity generation and consumption data and record them digitally. These readings can be called from the SCADA system, but the Project owner has no means to manipulate the readings or the measurement devices, as they will be sealed by TEIAS. On the first days of each successive month, power generation and consumption readings from both measurement devices will be read by TEIAS and a monthly protocol will be prepared based on these readings. This protocol will later be uploaded to the PMUM web site, where the Project owner can easily access and read daily and monthly power reports.

The electricity generation and consumption data that are recorded and reported during the monitoring period will be compiled with a spreadsheet program for verification purposes. This spreadsheet, along with documentary evidence, will be provided to the verifying DOE for verification. The sample monitoring spreadsheet is also provided to the validating DOE.

The following procedure will be used to calculate the emission reductions:

The measurement instruments give two types of data (beside others); monthly gross electricity generated and monthly electricity consumed by the Project. The difference of these two parameters is the monthly net electricity supplied to the grid. These monthly net electricity supply figures will be added up for all months during the monitoring period and the sum will be multiplied with the ex-ante CM emission factor to find the amount of emissions reductions. All the data will

be electronically archived and communicated, and the calculations will be reported in a separate spreadsheet for verification purposes. The calculations will be automatically done by the spreadsheet file with minimal human interaction. The formulae that will be used for monitoring are as the following:

$$EG_{\text{facility},y,m} = EG_{\text{facility,gross},y,m} - EG_{\text{facility,net},y,m} \quad [7]$$

$$ER_y = \sum_{y,m} EG_{\text{facility},y,m} \quad [8]$$

Table 23. Explanations of parameters used for calculations

Parameter	Explanation
$EG_{\text{facility},y,m}$	Quantity of net electricity supplied to grid by the Project plant in year y and month m (kWh)
$EG_{\text{facility,gross},y,m}$	Quantity of gross electricity supplied to grid by the Project plant in year y and month m (kWh). In the PMUM reports in Turkish, this corresponds to "ISVM".
$EG_{\text{facility,net},y,m}$	Quantity of electricity consumed from grid by the Project plant in year y and month m (kWh). In the PMUM reports in Turkish, this corresponds to "UECM".
ER_y	Emission reductions in year y (tCO ₂)
y	Year of monitoring period
m	Month of monitoring period

Source of Data:

The primary source of data is the PMUM web site³⁶, which is the basis for financial settlement and invoicing. The Project owner has an access code for its own projects to the PMUM web site and can call, report and print actual and historical electricity generation and consumption data of the Project activity. These PMUM reports can only be obtained by the Project owner, therefore screenshots will also be provided during verification for QA/QC reasons.

In case there is any unforeseen problem with the PMUM data, the secondary source of data are the invoices. In this case, the invoices shall be cross checked with the monthly measurement protocols, which are read from the primary and secondary electricity meters by TEIAS and signed by TEIAS and the Project owner's responsible person.

If any other unforeseen problem occurs, the electricity generation and consumption data used for invoicing or settlement with TEIAS will be valid.

Frequency of Monitoring and Reporting:

The measurement devices measure and record data continuously and these readings can be monitored online by the Project owner's SCADA system. These data will be collected and reported once a month by TEIAS, and this measurement protocol is uploaded to the PMUM web site for invoicing purposes. The monitoring of emission reductions will also be carried out on a monthly basis, parallel to electricity generation reporting. Therefore, the frequency of monitoring is continuous and the frequency of reporting is monthly. The collected data will also be kept by the Project owner during and at least two years after the credit period.

³⁶ Source: <http://dgpys.teias.gov.tr>

Roles and Responsibilities:

The carbon manager of the Project owner is responsible for the preparation of the monitoring report. The Project owner is responsible for assigning a personnel as the carbon manager, whose tasks include:

- Collecting and archiving monthly PMUM reports and screenshots electronically and as hardcopy,
- Collecting and archiving monthly invoices and TEIAS protocols electronically and as hardcopy,
- Reporting in written form and collecting documentary evidence about any unusual issue regarding protocols, invoices, PMUM web site access, settlements, technical problems that might affect data quality and availability, significant equipment repair or maintenance and calibration, failure or replacement of measurement devices.
- Ensuring that information flow about the issues mentioned above between the Project manager on site and the carbon manager runs continuously during the whole monitoring period.
- Prepare the monitoring report, or provide the collected data to any assigned 3rd party that is responsible for supporting the verification of emission reductions.

Non-Conformities:

Before installation, metering devices are technically checked, calibrated and certified by independent companies for their accuracy. Afterwards, TEIAS also tests both metering devices in order to ensure that the devices satisfy all requirements and are calibrated properly. In case of failure of the electricity meters or if any difference occurs between primary and secondary device TEIAS has to be informed by the Project owner for necessary maintenance and calibration. An agreement between the project participant and TEIAS will ensure that in case of problems or failures of the meters TEIAS will react as fast as possible to solve the problem and perform necessary maintenance and calibration. Since the Project owner will invoice TEIAS for the electricity it sells to the grid, the data is of high quality. If it is determined by TEIAS that the seal of the primary metering device is torn apart or any intervention is made; correct power amounts are determined over the secondary meter, beginning from the latest measurement when the recorded values of the primary and the secondary metering devices match. If any difference is seen by the Project owner between primary and secondary devices, TEIAS will be informed accordingly.

Calibration of measurement devices is the responsibility of TEIAS and it is managed by the same authority, without Project owner's control according to the current legislation. If the responsibilities and legal requirements about calibration would change, the Project owner will act accordingly.

Since monitoring of the electricity generation and consumption is in the responsibility of TEIAS, the Project owner has little control over how, where and when TEIAS makes the readings. In case the electricity data cannot or does not get recorded by TEIAS on site, but rather on the nearest transformer station for any unknown reason (e.g. insufficient road access to the Project due to weather conditions etc., data transmission problem etc.), protocols might not be available. In these cases again the PMUM electricity data (as on the PMUM web site) will be valid.

Under normal circumstances, the protocols are signed monthly on site by TEIAS and the plant manager. After the protocol, internal SCADA data are cross checked with the protocol for quality control purposes. Usually information in both sources should be in agreement. However, should the data from both sources indicate a mismatch for any reason, then the Project owner may apply to TEIAS for correction of data and financial settlement. These cases will be reported in the monitoring report and if any settlement takes place the final figures will be used for calculations and documentary evidence will be provided to the verifying DOE.

Other non-conformities regarding readings, calibration and other TEIAS or PMUM system related

issues are subject to the bilateral communication and settlements between TEIAS and the Project owner. In such a case, the results of these settlements will be used as the valid data source.

5 ENVIRONMENTAL IMPACT

In agreement with national legislation³⁷, no full-EIA has been conducted for the Project. The Project Activity contributes positively towards sustainable development in Turkey in terms of social, environmental, technological and economical well-being:

- Power shortages are common in Turkey, especially during peak demand due to insufficient capacity. With the Project Activity, a small but positive contribution is made to the grid stability and quality of electricity,
- Reduction of electricity imports of Turkey, dependency on fossil fuels and increased energy security and lower supply risks and improvement in price stability,
- Diversification of energy resources and support to development of the renewable energy sector in Turkey,
- Reduction of greenhouse gas emissions (e.g. carbon dioxide) and air pollutants (e.g. particulates, sulphur dioxide, nitrogen oxides etc.) by displacing electricity from fossil fuel based power plants,
- Encouraging investments in the region and creation of new jobs in the region during construction and operation phases and support to local economy by procuring available products and services from the region,
- Contribution to a more sustainable energy generation infrastructure by showing that small run-of-river type hydropower projects can bring a high contribution to local sustainable development.

The EIA exemption approval has been obtained as of 17.12.2009.

³⁷ According to the new national EIA regulation, run-of-river type HEPP projects are defined as projects with a reservoir area less than 15 km² or with a reservoir volume less than 100 million m³. A full-EIA Report and a positive confirmation are required for run-of-river type HEPP projects larger than 50 MW installed capacity. For projects between 10 – 50 MW, some criteria are applied to assess whether an EIA is necessary or not. Projects under 10 MW power are exempt from full-EIA requirements. All projects need to prepare and submit a Project Introduction Report (a pre-EIA) to the EIA Directorate in the Ministry of Environment. For the transmission line, a separate EIA for the electricity line is required if the voltage of the transmission line is 154 kV or above.

6 STAKEHOLDER COMMENTS

During planning and construction phases, local stakeholders have been contacted by the Project owner to inform them about the Project and to gather their comments.

Land has been acquired either by expropriating land in cases the land owners could not be reached, or by settling with land owners on a willing buyer-willing seller relationship. The Project has not resulted in any involuntary resettlement. Most of the land is owned by public, which is provided to the Project owner.

The Project owner plans to employ several local people during construction and operation of the Project.

7 ANNEXES

Table 24. Capacity additions selected for BM calculations²²

POWER PLANT	CAPACITY [MW]	GENERATION [GWh]	FUEL TYPE
ÇAN GR I	160.0	1040.0	Lignite
ÇAN GR II	160.0	1040.0	Lignite
ELBİSTAN-B GR I	360.0	2340.0	Lignite
AKÇA ENERJİ GR-III	8.7	65.4	Natural Gas
AYKA TEKSTİL GR-I	5.5	40.0	Natural Gas
BAYDEMİRLER GR IV-V-VI	6.2	51.4	Natural Gas
BOSEN GR-III	50.0	350.0	Natural Gas
ÇUMRA ŞEKER	16.0	40.0	Natural Gas
ETİ MAD.(BAN.ASİT)GR-I	11.5	85.0	Renewables
EVYAP GR I-II	5.1	30.0	Natural Gas
GRANİSER GRANİT GR-I	5.5	42.0	Natural Gas
HABAŞ ALİAĞA GR III	47.7	381.6	Natural Gas
HABAŞ ALİAĞA GR IV	47.7	381.6	Natural Gas
HABAŞ ALİAĞA GR-V	24.6	196.8	Natural Gas
HAYAT KAĞIT GR-I	7.5	56.0	Natural Gas
İÇDAŞ ÇELİK GR-I	135.0	1080.0	Coal
KAHRAMANMARAŞ KAĞIT GR-I	6.0	45.0	Coal
KORUMA KLOR GR I-II-III	9.6	77.0	Natural Gas
KÜÇÜKÇALIK TEKSTİL GR I-II-III-IV	8.0	64.0	Natural Gas
MERCEDES BENZ TURK GR I-II-III-IV	8.3	68.0	Natural Gas
MODERN ENERJİ GR-III	8.4	62.9	Natural Gas
MODERN ENERJİ GR-II	6.7	50.4	Natural Gas
MOSB GR I-II-III-IV-V-VI-VII	84.8	434.0	Natural Gas
ORS RULMAN	12.4	99.4	Natural Gas
PAK GIDA(Kemalpaşa) GR-I	5.7	45.0	Natural Gas
TEZCAN GALVANİZ GR I-II	3.7	29.0	Natural Gas
YONGAPAN(KAST.ENTG) GR-II	5.2	32.7	Natural Gas
ZEYNEP GİYİM SAN. GR-I	1.2	9.0	Natural Gas
AK ENERJİ(K.paşa) GR- III	40.0	256.9	Natural Gas
AK ENERJİ(K.paşa) GR I-II	87.2	560.1	Natural Gas
ALTEK ALARKO GR I-II	60.1	420.0	Natural Gas
BİS ENERJİ GR VII	43.7	360.8	Natural Gas
CAN ENERJİ GR-I	3.9	28.0	Natural Gas
ÇEBİ ENERJİ BT	21.0	164.9	Natural Gas
ÇEBİ ENERJİ GT	43.4	340.1	Natural Gas
ENTEK ELK.A.Ş.KOÇ ÜNİ. GR I-II	2.3	19.0	Natural Gas
KAREGE GR IV-V	18.1	141.9	Natural Gas
KARKEY(SİLOPİ-4) GR-IV	6.2	47.2	Fuel Oil
KARKEY(SİLOPİ-4) GR-V	6.8	51.9	Fuel Oil
METEM ENERJİ(Hacışiramat) GR I-II	7.8	58.0	Natural Gas
METEM ENERJİ(Peliklik) GR I-II-III	11.7	89.0	Natural Gas
NOREN ENERJİ GR-I	8.7	70.0	Natural Gas
NUH ENERJİ-2 GR I	47.0	319.7	Natural Gas
ZORLU ENERJİ KAYSERİ GR-I-II-III	149.9	1144.1	Natural Gas
ZORLU ENERJİ KAYSERİ GR-IV	38.6	294.9	Natural Gas
ZORLU ENERJİ YALOVA GR I-II	15.9	122.0	Natural Gas
İÇTAŞ ENERJİ(Yukarı Mercan) GR I-II	14.2	44.0	Hydro

MURATLI GR I-II	115.0	444.0	Hydro
BEREKET EN.(DALAMAN) GR XIII-XIV-XV	7.5	35.8	Hydro
YAMULA GRUP I-II	100.0	422.0	Hydro
SUNJÜT(RES) GR I-II	1.2	2.4	Geothermal & Wind
EKOTEN TEKSTİL GR-I	1.9	14.0	Natural Gas
ERAK GİYİM GR-I	1.4	10.0	Natural Gas
ALARKO ALTEK GR-III	21.9	158.3	Natural Gas
AYDIN ÖRME GR-I	7.5	60.0	Natural Gas
NUH ENERJİ-2 GR II	26.1	180.1	Natural Gas
MARMARA ELEKTRİK (Çorlu) GR I	8.7	63.0	Natural Gas
MARMARA PAMUK (Çorlu) GR I	8.7	63.0	Natural Gas
ENTEK (Köseköy) GR IV	47.6	378.2	Natural Gas
ELSE TEKSTİL (Çorlu) GR I - II	3.2	25.0	Natural Gas
SÖNMEZ ELEKTRİK (Çorlu) GR I - II	17.5	126.0	Natural Gas
MENDERES ELEKTRİK GR I	8.0	56.0	Geothermal & Wind
KASTAMONU ENTEGRE (Balıkesir) GR I	7.5	54.0	Natural Gas
BOZ ENERJİ GR I	8.7	70.0	Natural Gas
ADANA ATIK SU ARITMA TESİSİ	0.8	6.0	Renewables
AMYLUM NİŞASTA (ADANA)	14.3	34.0	Natural Gas
ŞIK MAKAS (Çorlu) GR I	1.6	13.0	Natural Gas
ELBİSTAN B GR III	360.0	2340.0	Lignite
ANTALYA ENERJİ GR I - II - III - IV	34.9	245.0	Natural Gas
HAYAT TEM. VE SAĞLIK GR I - II	15.0	108.0	Natural Gas
EKOLOJİK EN. (Kemerburgaz) GR I	1.0	6.0	Renewables
EROĞLU GİYİM (Çorlu) GR I	1.2	9.0	Natural Gas
CAM İŞ ELEKTRİK (Mersin) GR I	126.1	1008.0	Natural Gas
ELBİSTAN B GR II	360.0	2340.0	Lignite
YILDIZ ENT. AĞAÇ (Kocaeli) GR I	6.2	40.0	Natural Gas
ÇERKEZKÖY ENERJİ GR I	49.2	390.0	Natural Gas
ENTEK (Köseköy) GR V	37.0	293.9	Natural Gas
ELBİSTAN B GR IV	360.0	2340.0	Lignite
ÇIRAĞAN SARAYI GR I	1.3	11.0	Natural Gas
AKMAYA (Lüleburgaz) GR I	6.9	50.0	Natural Gas
BURGAZ (Lüleburgaz) GR I	6.9	54.0	Natural Gas
SEYHAN I-II	0.3	1.7	Hydro
ŞANLIURFA GR I-II	51.8	124.0	Hydro
BEREKET ENERJİ GÖKYAR HES 3 Grup	11.6	43.3	Hydro
MOLU EN. Zamanlı Bahçelik GR I - II	4.2	16.7	Hydro
SU ENERJİ (Balıkesir) GR I - II	4.6	20.7	Hydro
BEREKET EN.(Mentaş Reg) GR I - II	26.6	108.7	Hydro
EKİN (Başaran Hes) (Nazilli)	0.6	4.5	Hydro
ERE(Sugözü rg. Kızıldüz hes) GR I - II	15.4	31.6	Hydro
ERE(AKSU REG.ve ŞAHMALLAR HES) GR I-II	14.0	26.7	Hydro
BEREKET EN.(Mentaş Reg) GR III	13.3	54.4	Hydro
HABAŞ (Aliağa-ilave)	9.1	62.0	Natural Gas
MODERN ENERJİ	5.2	35.5	Natural Gas
ARENKO	0.7	4.8	Natural Gas
ALTINMARKA GIDA	0.1	0.7	Natural Gas
TEKBOY ENERJİ	0.1	0.7	Natural Gas
VELSAN AKRİLİK	0.1	0.7	Natural Gas
Acıbadem Sağlık Hiz.ve Tic.A.Ş.(Kadıköy Hast.)(İstanbul/Kadıköy)	0.5	3.4	Natural Gas
Acıbadem Sağlık Hiz.ve Tic.A.Ş.(Kozyatağı Hast.)(İstanbul/Kadıköy)	0.6	4.1	Natural Gas
Acıbadem Sağlık Hiz.ve Tic.A.Ş.(Nilüfer/BURSA)	1.3	8.9	Natural Gas

AKATEKS Tekstil Sanayi ve Ticaret A.Ş.	1.8	12.3	Natural Gas
FLOKSER TEKSTİL SAN.AŞ.(Çatalca/İstanbul)(Poliser Tesisi)	2.1	14.3	Natural Gas
FLOKSER TEKSTİL SAN.AŞ.(Çatalca/İstanbul)(Sütser Tesisi)	2.1	14.3	Natural Gas
FRITOLAY GIDA SAN.VE TİC. AŞ.	0.5	3.4	Natural Gas
KIVANÇ TEKSTİL SAN.VE TİC.A.Ş.	3.9	26.6	Natural Gas
KİL-SAN KİL SAN.VE TİC. A.Ş	3.2	21.8	Natural Gas
SÜPERBOY BOYA SAN.ve Tic.Ltd.Şti.(Büyüçekmece/İstanbul)	1.0	6.8	Natural Gas
SWISS OTEL(Anadolu Japan Turizm A.Ş (İstanbul)	1.6	10.9	Natural Gas
TAV Esenboğa Yatırım Yapım ve İşletme AŞ./ANKARA	3.9	26.6	Natural Gas
NUH ENERJİ-2(Nuh Çim.)	73.0	497.7	Natural Gas
AKTEKS	0.8	5.9	Fuel Oil
UŞAK ŞEKER (NURİ ŞEKER)	1.7	12.8	Lignite
BOĞAZLIYAN ŞEKER	16.4	111.8	Natural Gas
KARTONSAN	5.0	34.1	Natural Gas
ESKİŞEHİR END.ENERJİ	3.5	23.9	Natural Gas
ESKİŞEHİR ŞEKER (KAZIM TAŞKENT)	2.9	19.8	Natural Gas
İGSAŞ	2.2	15.0	Natural Gas
DESA	0.7	4.8	Natural Gas
DENTAŞ	0.3	2.0	Natural Gas
SÜPER FILMCİLİK	0.1	0.7	Natural Gas
ATAER ENERJİ	0.1	0.7	Natural Gas
BİL ENERJİ	0.1	0.7	Natural Gas
BİS Enerji Üretim AŞ.(Bursa)(İlave)	43.0	293.2	Natural Gas
Aliağa Çakmaktepe Enerji A.Ş.(Aliağa/İZMİR)	34.8	237.3	Natural Gas
BİS Enerji Üretim AŞ.(Bursa)(İlave)	48.0	327.3	Natural Gas
BOSEN ENERJİ ELEKTRİK AŞ.	142.8	973.6	Natural Gas
SAYENERJİ ELEKTRİK ÜRETİM AŞ. (Kayseri/OSB)	5.9	40.2	Natural Gas
T ENERJİ ÜRETİM AŞ.(İSTANBUL)	1.6	10.9	Natural Gas
ZORLU EN.Kayseri (İlave 1 GT)	7.2	49.1	Natural Gas
SIİRT	25.6	190.0	Fuel Oil
Mardin Kızıltepe	34.1	250.0	Fuel Oil
KAREN	24.3	180.0	Fuel Oil
İDİL 2 (PS3 A- 2)	24.4	180.0	Fuel Oil
BORÇKA HES	300.6	1203.1	Hydro
YPM Ener.Yat.AŞ.(Altıntepe Hidro.)(Sivas/Suşehir)	4.0	16.0	Hydro
YPM Ener.Yat.AŞ.(Beyyınar Hidro.)(Sivas/Suşehir)	3.6	14.4	Hydro
YPM Ener.Yat.AŞ.(Konak Hidro.)(Sivas/Suşehir)	4.0	16.0	Hydro
KURTEKS Tekstil A.Ş./Kahramanmaraş(KARASU HES-Andırın)	2.4	9.6	Hydro
İSKUR TEKSTİL (SÜLEYMANLI HES)	4.6	18.4	Hydro
ÖZGÜR ELK.AŞ.(K.MARAŞ)(Tahta)	6.3	25.2	Hydro
ÖZGÜR ELK.AŞ.(K.MARAŞ)(Tahta)(İlave)	6.3	25.2	Hydro
MB ŞEKER NİŞASTA SAN. A.Ş. (Sultanhanı)	8.8	60.0	Natural Gas
AKSA ENERJİ (Antalya)	183.8	1290.0	Natural Gas
AKSA ENERJİ (Manisa)	52.4	370.0	Natural Gas
ANTALYA ENERJİ (İlave)	17.5	122.3	Natural Gas
ATAÇ İNŞAAT SAN. A.S.B. (ANTALYA)	5.4	37.0	Natural Gas
BAHÇIVAN GIDA(LÜLEBURGAZ)	1.2	8.0	Natural Gas
CAN ENERJİ (Çorlu-TEKİRDAĞ)(İlave)	52.4	304.2	Natural Gas
FOUR SEASONS OTEL (ATIK PASHA)	1.2	7.0	Natural Gas

TUR.A.Ş.)			
FRITOLAY GIDA SAN. VE TİC. A.Ş. (İlave)	0.1	4.0	Natural Gas
KARKEY (SİLOPİ-5)(154kV)(İlave)	14.8	103.2	Fuel Oil
MELİKE TEKSTİL (GAZİANTEP)	1.6	11.0	Natural Gas
MİSİS APRE TEKSTİL BOYA EN. SAN.	2.0	14.0	Natural Gas
MODERN ENERJİ (LÜLEBURGAZ)	13.4	94.1	Natural Gas
ORTADOĞU ENERJİ (ODA YERİ)(Eyüp-İst)	2.8	22.0	Renewables
POLAT TURZ. (POLAT RENAISSANCE İST.OT.)	1.6	11.0	Natural Gas
SARAYKÖY JEOTERMAL(DENİZLİ)	6.9	50.0	Geothermal & Wind
YILDIZ SUNTA (Uzunçiftlik-Köseköy)(Düzeltilme)	22.6	146.5	Natural Gas
SÖNMEZ Elektrik (İlave)	8.7	67.3	Natural Gas
AKKÖY ENERJİ (AKKÖY I HES)	101.9	408.0	Hydro
ALP ELEKTRİK (TINAZTEPE) ANTALYA	7.7	29.0	Hydro
CANSU ELEKTRİK (Murgul/ARTVİN)	9.2	47.0	Hydro
DAREN HES ELKT. (SEYRANTEPE BARAJI VE HES)	49.7	182.0	Hydro
DEĞİRMENÜSTÜ EN. (KAHRAMANMARAŞ)	25.7	69.0	Hydro
GÖZEDE HES (TEMSA ELEKTRİK) BURSA	2.4	10.0	Hydro
H.G.M. ENERJİ (KEKLİCEK HES) (Yeşilyurt)	8.7	18.0	Hydro
HİDRO KNT.(YUKARI MANAHOZ REG.VE HES)	22.4	79.0	Hydro
İÇ-EN ELK.(ÇALKIŞLA REGÜLATÖRÜ VE HES)	7.7	18.0	Hydro
KALEN ENERJİ (KALEN II REGÜLAT. VE HES)	15.7	50.0	Hydro
MARAŞ ENERJİ (FIRNIS REGÜLATÖRÜ VE HES)	7.2	36.0	Hydro
SARMAŞIK I HES (FETAŞ FETHİYE ENERJİ)	21.0	96.0	Hydro
SARMAŞIK II HES (FETAŞ FETHİYE ENERJİ)	21.6	108.0	Hydro
TORUL	105.6	322.0	Hydro
YEŞİL ENERJİ ELEKTRİK (TAYFUN HES)	0.8	5.0	Hydro
AK GIDA SAN. VE TİC. A.Ş. (Pamukova)	7.5	61.0	Natural Gas
AKSA AKRİLİK KİMYA SN. A.Ş. (YALOVA)	70.0	539.0	Natural Gas
AKUA ENERJİ (KAYALIK REG. VE HES)	5.8	39.0	Hydro
ALKİM ALKALİ KİMYA (Cihanbeyli/KONYA)	0.4	3.0	Lignite
ANADOLU ELEKTRİK (ÇAKIRLAR HES)	16.2	60.0	Hydro
ARENKO ELEKTRİK ÜRETİM A.Ş. (Denizli)	12.0	84.0	Natural Gas
BAĞISLI REG. VE HES (CEYKAR ELEKT.)	9.9	33.1	Hydro
BAĞISLI REG. VE HES (CEYKAR ELEKT.)	19.7	65.9	Hydro
BEREKET ENERJİ (KOYULHİSAR RES)	42.0	329.0	Geothermal & Wind
CARGILL TARIM VE GIDA SAN. TİC. A.Ş.	0.1	1.1	Renewables
CİNDERE HES (Denizli)	19.1	58.0	Hydro
ÇELİKLER TAAH. İNŞ. (RİXOX GRAND)	2.0	16.0	Natural Gas
DALSAN ALÇI SAN. VE TİC. A.Ş.	1.2	9.0	Natural Gas
DELTA ENERJİ ÜRETİM VE TİC. A.Ş.	47.0	365.8	Natural Gas
DENİZLİ ELEKTRİK (EGE I HES)	0.9	4.0	Hydro
DESA ENERJİ ELEKTRİK ÜRETİM A.Ş.	9.8	70.0	Natural Gas
E.ŞEHİR END. ENERJİ (DG+M) (Eskişehir-2)	59.0	452.0	Natural Gas
EGE BİRLEŞİK ENERJİ (LPG+DG+M) (Aliğa)	12.8	107.0	Natural Gas
ERDEMİR (Ereğli-Zonguldak)	39.2	240.0	Natural Gas
ERVA ENERJİ (KABACA REG. VE HES)	4.2	16.3	Hydro
ERVA ENERJİ (KABACA REG. VE HES)	4.2	16.3	Hydro
FALEZ ELEKTRİK ÜRETİMİ A.Ş.	11.7	88.0	Natural Gas
GLOBAL ENERJİ (PELİTLİK)	8.6	65.7	Natural Gas
GÜL ENERJİ ELKT.ÜRET. SN. VE TİC. A.Ş.	24.3	170.0	Fuel Oil
GÜRMAT ELEKT. (GÜRMAT JEOTERMAL)	47.4	313.0	Geothermal & Wind
KALEN ENERJİ (KALEN I-II HES)	15.7	52.2	Hydro

KASAR DUAL TEKSTİL SAN. A.Ş. (Çorlu)	5.7	38.0	Natural Gas
KAYEN ALFA ENERJİ (KALETEPE HES)	10.2	37.0	Hydro
KEN KİPAŞ ELEKTRİK ÜRETİM (KAREN)	41.8	180.0	Natural Gas
KEN KİPAŞ ELKT. ÜR. (KAREN) (K.Maraş)	17.5	75.4	Fuel Oil
LAMAS III-IV HES (TGT ENERJİ ÜRETİM)	35.7	150.0	Hydro
MAURİ MAYA SAN. A.Ş.	0.3	2.5	Renewables
MAURİ MAYA SAN. A.Ş.	2.0	16.5	Renewables
ÖZGÜR ELEKTRİK (AZMAK II REG. VE HES)	24.4	91.0	Hydro
ÖZYAKUT ELEK. ÜR. A.Ş. (GÜNEŞLİ HES)	0.6	2.7	Hydro
ÖZYAKUT ELEK. ÜR. A.Ş. (GÜNEŞLİ HES)	1.2	5.3	Hydro
RASA ENERJİ (VAN)	78.6	500.0	Natural Gas
SARITEPE HES (GENEL DİNAMİK SİS. EL.)	2.5	10.2	Hydro
SARITEPE HES (GENEL DİNAMİK SİS. EL.)	2.5	10.2	Hydro
SELKASAN KAĞIT PAKETLEME MALZ. İM.	9.9	73.0	Natural Gas
SİLOPİ ELEKTRİK ÜRETİM A.Ş.	135.0	945.0	Fuel Oil
SİLOPİ ELEKTRİK ÜRETİM A.Ş. (ESENBOĞA)	44.8	315.0	Fuel Oil
TAŞOVA YENİDEREKÖY HES (HAMEKA A.Ş.)	2.0	10.0	Hydro
TAV İSTANBUL TERMİNAL İŞLETME A.Ş.	3.3	27.6	Natural Gas
TAV İSTANBUL TERMİNAL İŞLETME A.Ş.	6.5	54.4	Natural Gas
TESKO KİPA KİTLE PAZ. TİC. VE GIDA A.Ş.	2.3	18.0	Natural Gas
TOCAK I HES (YURT ENERJİ ÜRETİM SN.)	4.8	13.0	Hydro
UZUNÇAYIR HES (Tunceli)	27.3	105.0	Hydro
YURTBAY ELEKTRİK ÜRETİM A.Ş. (D.G.+M)	6.9	50.0	Natural Gas